

PRACTICAL—2

Name:- Neha guglot

PRN:- 202501050032

2.1.1. List operations

Write a Python program that implements a menu-driven interface for managing a list of integers. The program should have the following menu options:

1. Add
2. Remove
3. Display
4. Quit

The program should repeatedly prompt the user to enter a choice from the menu. Depending on the choice selected, the program should perform the following actions:

- **Add:** Prompts the user to enter an integer and add it to the integer list. If the input is not a valid integer, display "Invalid input".
- **Remove:** Prompts the user to enter an integer to remove from the list. If the integer is found in the list, remove it; otherwise, display "Element not found". If the list is empty, display "List is empty".
- **Display:** Displays the current list of integers. If the list is empty, display "List is empty".
- **Quit:** Exits the program.
- The program should handle invalid menu choices by displaying "Invalid choice". Ensure that the program continues to prompt the user until they choose to quit (option 4).

Code:

```
l1=[]  
while True:  
    print("1. Add")  
    print("2. Remove")  
    print("3. Display")  
    print("4. Quit")  
    n = int(input("Enter choice: "))  
    if n == 1:  
        add = int(input("Integer: "))
```

```
l1.append(add)

print(f"List after adding: {l1}")

elif n == 2:

    if len(l1) == 0:

        print("List is empty")

    elif len(l1) != 0:

        remove=int(input("Integer: "))

        if remove not in l1 :

            print("Element not found")

        else:

            l1.remove(remove)

            print(f"List after removing: {l1}")

elif n==3:

    if len(l1) ==0:

        print("List is empty")

    else:

        print(l1)

elif n==4:

    break

else:

    print("Invalid choice")
```

The image displays two screenshots of the CodeTANTRA web interface, showing test results for a Python program named `listOps.py`.

Top Screenshot: Test case 1

- Performance:** Average time 0.070 s (70.00 ms), Maximum time 0.106 s (106.00 ms). 2 out of 2 shown test case(s) passed, 1 out of 1 hidden test case(s) passed.
- Expected output:**

```

1. Add
2. Remove
3. Display
4. Quit
Enter choice: 1
Integer: 11
List after adding: [11]
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 1
Integer: 25
List after adding: [11, 25]
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 2
Integer: 26
Element not found
1. Add

```
- Actual output:**

```

1. Add
2. Remove
3. Display
4. Quit
Enter choice: 1
Integer: 11
List after adding: [11]
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 1
Integer: 25
List after adding: [11, 25]
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 2
Integer: 26
Element not found
1. Add

```

Bottom Screenshot: Test case 2

- Performance:** Average time 0.070 s (70.00 ms), Maximum time 0.106 s (106.00 ms). 2 out of 2 shown test case(s) passed, 1 out of 1 hidden test case(s) passed.
- Expected output:**

```

1. Add
2. Remove
3. Display
4. Quit
Enter choice: 3
List is empty
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 2
List is empty
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 3
Invalid choice
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 4

```
- Actual output:**

```

1. Add
2. Remove
3. Display
4. Quit
Enter choice: 3
List is empty
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 2
List is empty
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 3
Invalid choice
1. Add
2. Remove
3. Display
4. Quit
Enter choice: 4

```

2.1.2. Dictionary Operations

Write a Python program to perform insertion, update, deletion, and traversal operations on a dictionary. An initial dictionary containing 10 predefined records is already given in the program.

Operations to be Performed:

1. Insertion – Insert a new key-value pair into the dictionary using user input.
2. Update – Update the value of an existing key using user input.
3. Deletion – Delete a specified key from the dictionary using user input.
4. Traversal – Traverse the final dictionary and display all key-value pairs.

Input and Output Format:

1. Read an integer representing the key to be inserted and a string representing its value. Insert this new key-value pair into the dictionary and display the dictionary after insertion.

2. Read an integer representing the key to be updated and a string representing the new value. Update the value of the specified key (only if it exists) and display the dictionary after the update.
3. Read an integer representing the key to be deleted. Delete the specified key from the dictionary (only if it exists) and display the dictionary after deletion.
4. Finally, traverse the dictionary and display all key-value pairs.

The program should also display the original dictionary before performing any operations. Each output should be printed with appropriate messages.

Note:

- All operations must be performed using dictionary methods.
- Perform deletion only if possible; leave the dictionary unchanged.
- Refer to the visible test cases for better understanding and strictly match with the input/outputs.

Code:

```
# Initial dictionary with 10 predefined records
```

```
student = {  
    1: "Amit",  
    2: "Riya",  
    3: "Kiran",  
    4: "Neha",  
    5: "Arjun",  
    6: "Pooja",  
    7: "Rahul",  
    8: "Sneha",  
    9: "Vikram",  
    10: "Anjali"  
}
```

```
# write your code here...
```

```
print("Original Dictionary:",student)
```

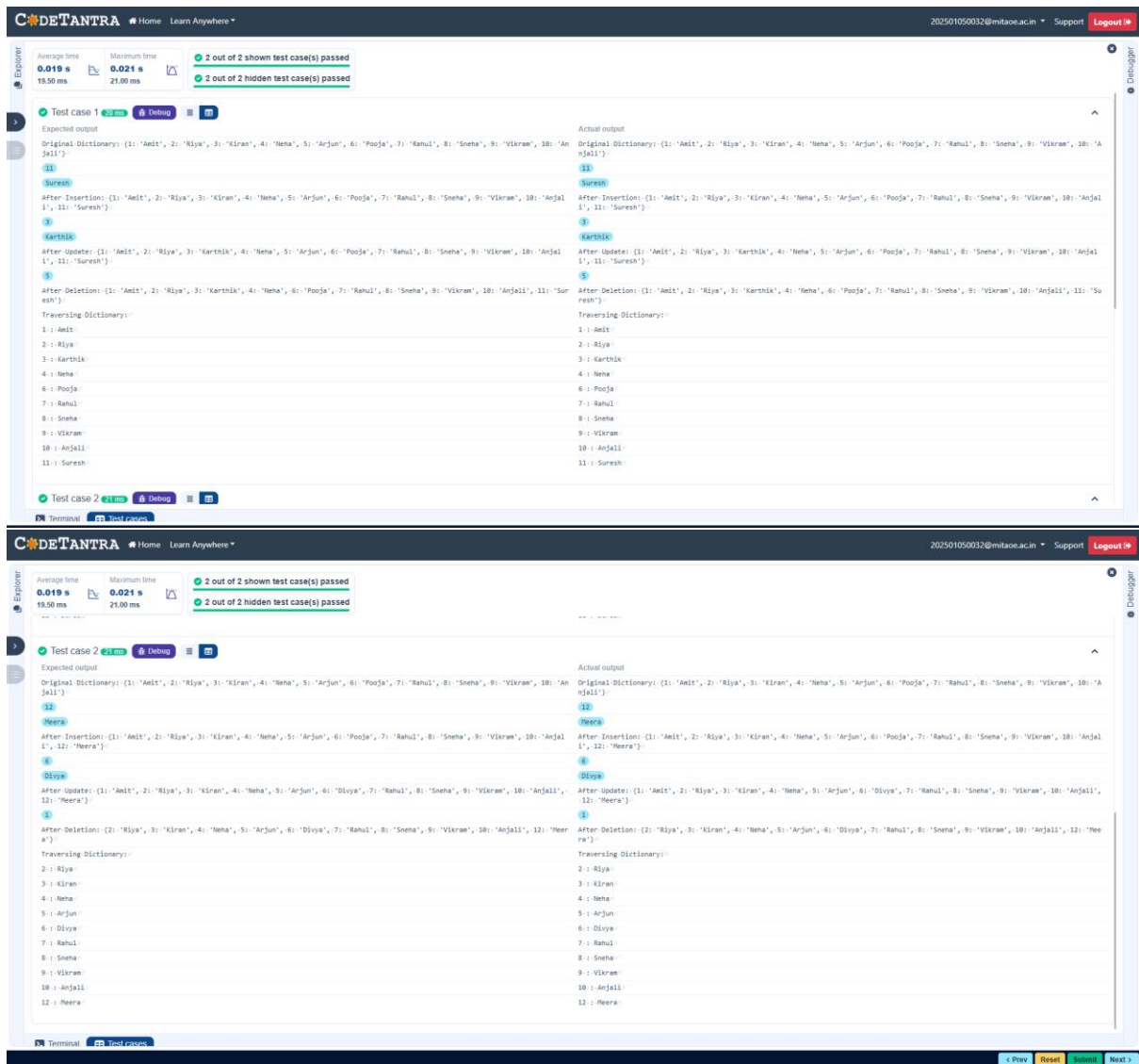
```
key1=int(input())
```

```
value1=input()
student.update({key1:value1})
print("After Insertion:",student)
```

```
key2=int(input())
value2=input()
if(key2 in student.keys()):
    student.update({key2:value2})
print("After Update:",student)
```

```
key3=int(input())
if(key3 in student.keys()):
    student.pop(key3)
print("After Deletion:",student)
```

```
print("Traversing Dictionary:")
for key,value in student.items():
    print(f"{key} : {value}")
```



2.2.1. Linear search Technique

Write a program to check whether the given element is present or not in the array of elements using linear search.

Input format:

- The first line of input contains the array of integers which are separated by space
- The last line of input contains the key element to be searched

Output format:

- If the element is found, print the index. If the element found multiple times, print the starting index
- If the element is not found, print **Not found**.

Sample Test Case:

Input:

1 2 3 4 3 5 6

3

Output:

2

Code:

```
def linear_search(arr, key):  
    for i in range(len(arr)):  
        if arr[i]==key:  
            return i  
    return -1
```

```
arr = list(map(int, input().split()))
```

```
key = int(input())
```

```
result= linear_search(arr,key)
```

```
if result != -1:
```

```
    print(result)
```

```
else:
```

```
    print("Not found")
```

The screenshot displays the CodeTANTRA IDE interface. At the top, the header includes the CodeTANTRA logo, navigation links (Home, Learn Anywhere), a user email (202501050032@mitace.ac.in), and a Logout button. The main workspace shows a code editor with the provided Python code. Below the editor, a test results panel indicates that 2 out of 2 shown test cases passed, with a total of 2 out of 2 hidden test cases also passed. The panel details two test cases: Test case 1 and Test case 2. For Test case 1, the expected output is '1 2 3 4 3 5 6' and the actual output is '1 2 3 4 3 5 6', with a key of '3' and a result of '2'. For Test case 2, the expected output is '1 2 3 4 5 6 7 8 9 10' and the actual output is '1 2 3 4 5 6 7 8 9 10', with a key of '20' and a result of 'Not found'.

2.2.2. Captain of the Team

You are provided with the heights of 11 cricket players (in centimeters). Your task is to identify the tallest player, who will be selected as the captain of the team.

Input Format:

The first line of input will contain 11 integers, each representing the height of a player (in centimeters), each separated by a space.

Output Format

The output should be the height (in centimeters) of the tallest player.

Code:

```
heights=list(map(int, input().split()))  
  
tallest_player = max(heights)  
  
print(tallest_player)
```

The screenshot displays the CodeTANTRA web-based IDE. The top navigation bar includes the logo, 'Home', 'Learn Anywhere', a user email '202501050032@mitase.ac.in', 'Support', and a 'Logout' button. The main workspace is divided into a left sidebar with an 'Explorer' view and a central editor area. The editor shows a Python script with three lines of code. Above the code, a status bar indicates '1 out of 1 shown test case(s) passed' and '2 out of 2 hidden test case(s) passed'. Below the code, a 'Test case 1' panel is open, showing 'Expected output' as '191' and 'Actual output' as '191', both highlighted in green. The right sidebar contains a 'Debugger' view.